Below is a **practical list of 50 sample AI risks** commonly identified when implementing **ISO/IEC 42001 (AI Management System – AIMS)** and performing **AI risk assessments aligned with ISO/IEC 23894**. These are grouped according to **major risk domains** typically used in AIMS risk registers.

1️⃣ Data Risks

1. Training data contains **demographic bias** leading to discriminatory outcomes.
2. **Incomplete datasets** leading to inaccurate model predictions.
3. **Outdated training data** causing incorrect results.
4. **Data labeling errors** affecting model performance.
5. **Lack of data provenance** (unknown data sources).
6. **Sensitive personal data used without consent**.
7. **Unbalanced datasets** leading to poor model fairness.
8. **Data poisoning attacks** manipulating training data.
9. **Inconsistent data formats** affecting model training.
10. **Poor data quality** leading to unreliable outputs.

2️⃣ Model Risks

11. **Model overfitting** resulting in poor real-world performance.
12. **Model underfitting** causing low prediction accuracy.
13. **Model drift** due to changing data patterns.
14. **Concept drift** affecting predictive accuracy over time.
15. **Black-box models lacking explainability**.
16. **Incorrect model assumptions**.
17. **Improper feature selection** affecting outcomes.
18. **Insufficient model validation** before deployment.
19. **Use of inappropriate algorithms** for the problem domain.
20. **Failure to retrain models regularly**.

3️⃣ Operational Risks

21. **Incorrect model version deployed in production.**
22. **Lack of monitoring of model performance.**
23. **Failure to detect declining accuracy.**
24. **Poor documentation of AI lifecycle processes.**
25. **Inadequate incident response for AI failures.**
26. **AI system not integrated properly with existing IT systems.**
27. **Uncontrolled model updates.**
28. **Insufficient change management procedures.**
29. **Inadequate testing before deployment.**
30. **Lack of defined AI system ownership.**

4️⃣ Security Risks

31. **Adversarial attacks on ML models.**
32. **Model extraction attacks via APIs.**
33. **Model inversion attacks revealing training data.**
34. **Unauthorized access to AI models.**
35. **Data leakage from AI training datasets.**
36. **Weak API security for AI services.**
37. **Malicious manipulation of AI outputs.**

5️⃣ Ethical and Societal Risks

38. **Algorithmic discrimination against certain groups.**
39. **Lack of transparency in AI decision-making.**
40. **Automation bias causing humans to trust AI blindly.**
41. **AI decisions affecting individuals without human oversight.**
42. **Use of AI in high-risk decisions without impact assessment.**
43. **AI-generated misinformation or harmful outputs.**

6️⃣ Governance and Compliance Risks

44. **Lack of documented AI governance framework.**
45. **Absence of AI risk management process.**
46. **Non-compliance with privacy regulations (GDPR, etc.).**
47. **Lack of internal audit coverage for AI systems.**
48. **Failure to maintain AI documentation and records.**

7️⃣ Third-Party and Supply Chain Risks

49. **Use of third-party AI models with unknown training data sources.**
50. **Dependence on external AI vendors without transparency or control.**